



Mini Project Report

on

CodeCase :

All round manager to your CP skills

For the course

Web Programming Laboratory

Submitted by

Om Thanage

Roll No: 16010123217

Omik Acharya

Roll No: 16010123218

Omkar Dinde

Roll No: 16010123219

Guide

Rohini Nair

Department of Computer Engineering

K. J. Somaiya School of Engineering

Somaiya Vidyavihar University

DECLARATION

I declare that the written Mini Project report submission represents the work done based on my and / or others' ideas with adequately cited and referenced the original source. I also declare that I have adhered to all principles of academic honesty and integrity as per norms of the Somaiya Vidyavihar University. I have not misinterpreted, fabricated, or falsified any idea/data/fact/source/original work/matter in my submission.

I understand that any violation of the above will be cause for disciplinary action by the university and may evoke the penal action from the sources which have not been properly cited or from whom proper permission is not sought.

Roll NO: 16010123217 Student Name: Om Thanage Signature	Roll NO: 16010123218 Student Name: Omik Acharya Signature
Roll NO: 16010123219 Student Name: Omkar Dinde Signature	

Date: 23/04/25

Place: Mumbai-77

Report Includes: This as on separate pages

1. Application Web Development: CodeCase
2. Literature Review about Application (With references)

Competitive Programming (CP) has emerged as a significant intellectual sport, fostering algorithmic thinking and problem-solving skills among programmers globally. Participants often engage with diverse online platforms, each offering unique challenges and communities (Halim & Halim, 2013). Recognizing the need for a consolidated experience, the "CodeCase" application provides an innovative solution, acting as a centralized hub for competitive programmers to streamline their training, track progress, and stay informed.

Core Functionality and Design

CodeCase is thoughtfully designed as a web application leveraging robust technologies like PHP and MySQL, complemented by a dynamic HTML/CSS/JavaScript front-end.

Technical Implementation and Integration

The application demonstrates effective use of modern web technologies and external APIs to deliver real-time, relevant data. It intelligently interfaces with established services to fetch contest schedules and user statistics, ensuring the information presented is current and accurate. This integration provides users with a seamless experience, eliminating the need to manually check multiple websites. The backend is built to efficiently handle data retrieval and processing, while the frontend focuses on delivering a clear, responsive, and engaging user interface.

Context and Innovation

While various tools exist to support aspects of competitive programming, such as contest aggregators or individual platform dashboards, CodeCase distinguishes itself by offering a holistic and integrated solution. It addresses the common challenge of fragmentation by bringing together contest tracking, multi-platform statistics, personalized problem and note management, and learning assistance into a single, cohesive platform. This user-centric approach aligns with the modern need for personalized, efficient dashboards that consolidate information and streamline workflows, specifically tailored for the vibrant competitive programming community.

References

.Halim, S., & Halim, F. (2013). *Competitive Programming 3: The New Lower Bound of Programming Contests*. Lulu Independent Publish.

3. Need Statement:

Competitive programming (CP) is a rapidly growing field that attracts numerous programmers seeking to enhance their algorithmic problem-solving skills. Participants typically engage with multiple online platforms such as LeetCode, CodeChef, and Codeforces, each offering distinct problem sets, contest formats, and community features. While this diverse ecosystem provides rich learning opportunities, it also presents significant challenges for programmers striving for consistent improvement.

Managing progress across these disparate platforms often requires manual tracking or navigating multiple websites, resulting in a fragmented view of overall performance and skill development. Staying informed about upcoming contests demands checking various schedules, increasing the risk of missing important events. Furthermore, learning resources, solved problems, and personal notes tend to be scattered, hindering efficient review and knowledge consolidation.

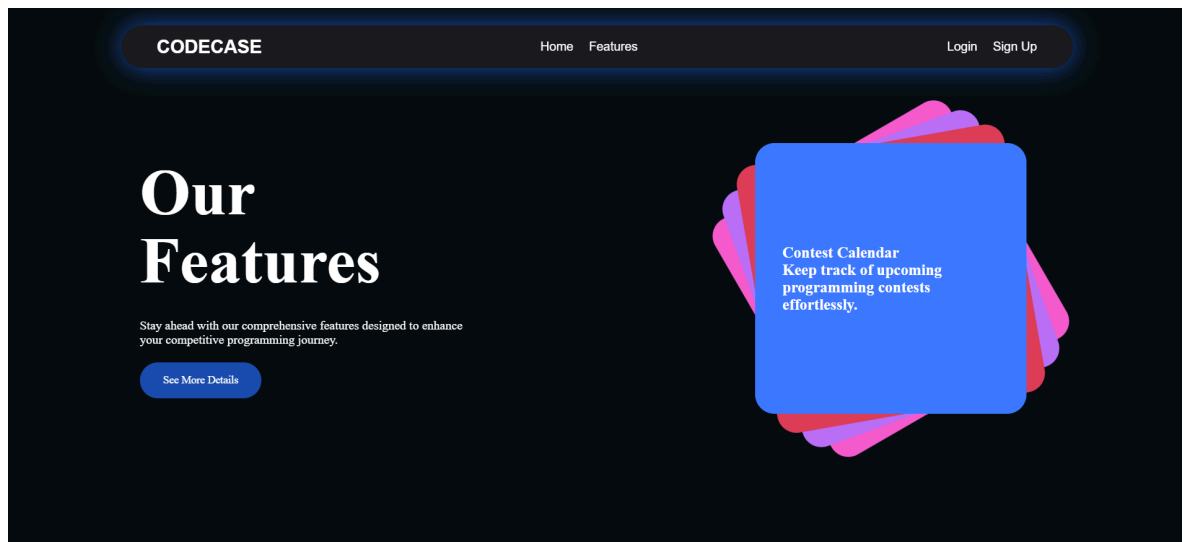
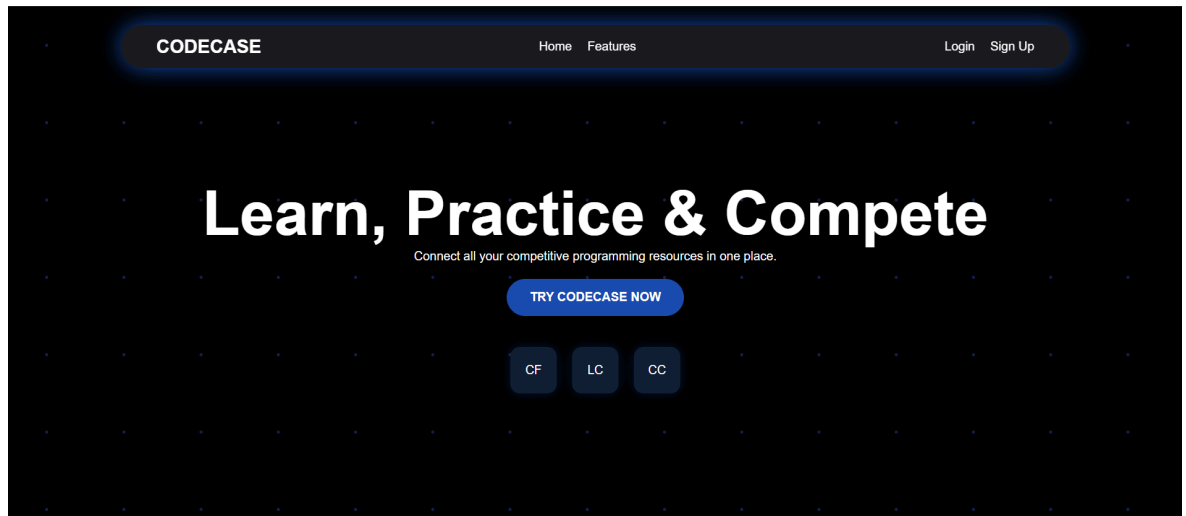
This fragmentation highlights the need for a centralized hub where competitive programmers can seamlessly integrate their activities. There is a clear demand for a tool that aggregates contest information, provides a unified view of performance statistics drawn from different platforms, facilitates the organization of learning materials, and offers convenient access to relevant resources. Such a platform would significantly streamline the training process, allowing programmers to focus more effectively on skill enhancement rather than administrative overhead.

4. Objectives:

- Centralized Contest management system
Integrate with LeetCode, CodeChef, and Codeforces APIs.
Implement contest calendar with IST timezone support.
Create functionality for users to save and track upcoming contests
- Problem Management System
Build a system to categorize and filter coding problems by platform and difficulty
Implement bookmarking functionality to save interesting problems
Create a solved problems tracking mechanism
- Personal notes
Developed a notes system for storing algorithms, solutions, and code snippets
Implemented organization tools for code snippets and programming concepts
- Progress Tracking
Integrate user statistics across multiple competitive programming platforms
- AI chatbot to provide hints to the queries asked
Implement the CP Helper feature to provide AI-powered assistance
Integrated support for competitive programming concepts and guidance

5. Flow of pages :

1) Landing page



2) Sign Up

The image shows a 'Create an account' form for a website called '<CodeCase>'. The form is centered on a dark background. It includes fields for 'Enter Name' (with the text 'Om Thanage'), 'Enter Email' (with the text 'om.thanage@somaiya.ec'), 'Password', and 'Confirm Password'. There is a checkbox for 'I accept the Terms and Conditions' which is checked. A blue button labeled 'Create an account' is at the bottom. Below the button, it says 'Already have an account? [Login](#)'.

Create an account <CodeCase>

Enter Name Enter Email

Om Thanage om.thanage@somaiya.ec

Password

Confirm Password

☒ I accept the Terms and Conditions

Create an account

Already have an account? [Login](#)

3) Login

The image shows a 'Login' form for a website called '<CodeCase>'. The form is centered on a dark background. It includes fields for 'Enter Email' (with the text 'om.thanage@somaiya.edu') and 'Password'. A blue button labeled 'Login' is at the bottom. Below the button, it says 'New here? [Sign Up](#)'.

Login <CodeCase>

Enter Email

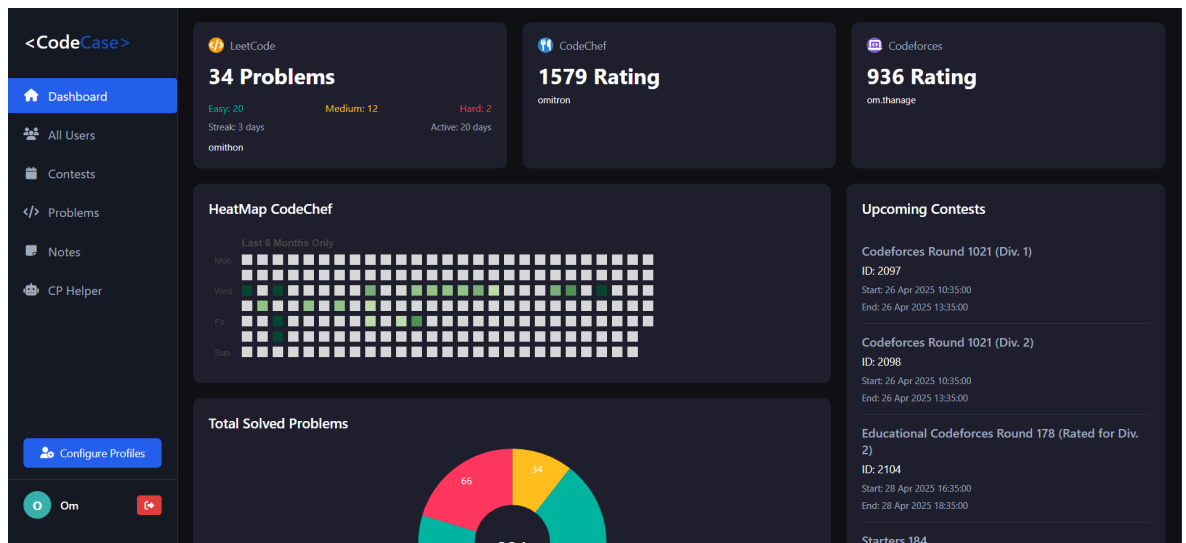
om.thanage@somaiya.edu

Password

Login

New here? [Sign Up](#)

4) Dashboard

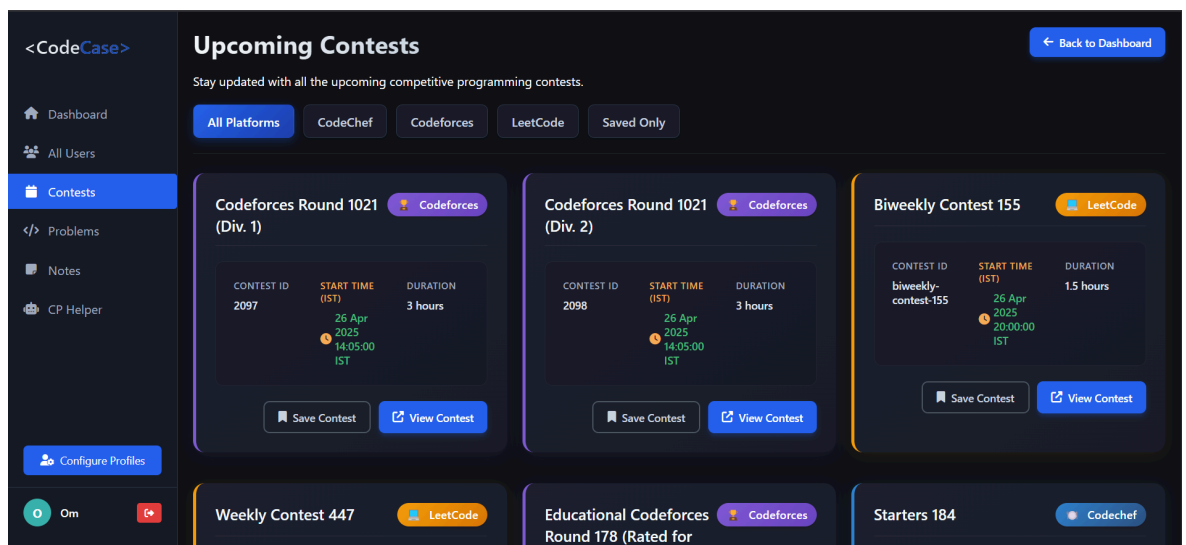


5) Profile

The profile page displays the 'Connect Profiles' form, which allows users to link their CodeChef, Codeforces, and LeetCode accounts. The form includes input fields for each platform's ID and an 'Update Profiles' button. A 'Back to Dashboard' link is also present.

Platform	ID
CodeChef	omithon
Codeforces	om.thanage
LeetCode	omithon

6) Contests Page



7) Problems Page

The screenshot shows the 'Coding Problems' page in the CodeCase application. The left sidebar contains navigation links: Dashboard, All Users, Contests, Problems (active), Notes, and CP Helper. Below these is a 'Configure Profiles' button and a user profile section for 'Om'. The main content area is titled 'Coding Problems' with a subtitle 'Browse, solve, and bookmark coding problems from various platforms.' and a 'Back to Dashboard' button. A search bar is at the top. Below it are filters for Platform (All, LeetCode, CodeChef, Codeforces, Custom), Difficulty (All, Easy, Medium, Hard), and Bookmarked (Bookmarked Only, All Problems). A 'Solved Status' section has 'Solved Only' and 'All Problems' buttons. Two problem cards are visible: 'Two Sum' (Easy, LeetCode) and 'Add Two Numbers' (Medium, LeetCode). Each card includes a description, tags (Array, Hash Table for Two Sum; Linked List, Math for Add Two Numbers), and buttons for Bookmark, Mark as Solved, and Solve.

8) Notes

The screenshot shows the 'Notes' page in the CodeCase application. The left sidebar is identical to the previous page, with 'Notes' now active. The main content area is titled 'Notes' with a subtitle 'Store and organize your programming notes, algorithms, and code snippets.' and a 'Back to Dashboard' button. On the left, there's an 'All Notes' section with a '+ New' button and a search bar. Below it are category counts: All Categories (3), Algorithms (2), and Programming (1). A list of notes is shown, including 'Dynamic Programming Patterns', 'Graph Traversal Techniques', and 'C++ STL Containers Cheat Sh...'. The 'Dynamic Programming Patterns' note is expanded, showing a list of 5 common DP patterns to remember: 1. Fibonacci sequence pattern, 2. 0/1 Knapsack pattern, 3. Unbounded Knapsack pattern, 4. Longest Common Subsequence pattern, and 5. Palindromic Subsequence pattern. The note also has a 'Category: Algorithms' tag and a 'Last updated: Apr 16, 2025 11:42' timestamp.

9) Cp_Helper (AI chatbot)

<CodeCase>

Dashboard

All Users

Contests

Problems

Notes

CP Helper

Configure Profiles

Om Thanage

CP Helper

Your AI assistant for competitive programming help and advice.

How do I improve my problem-solving skills?

What are some good resources for CP?

How to prepare for coding interviews?

What language is best for CP?

CP Helper Bot

Hello! I'm your CP Helper bot. How can I assist you with your competitive programming journey today?

Hey cp helper, give me topics I should revise to reach expert on codeforces!!!

To reach expert on Codeforces, focus on: 1. **Data Structures:** Advanced trees (segment, Fenwick), hash tables, sets/maps, priority queues. 2. **Algorithms:** Dynamic programming, graph algorithms (shortest paths, MST, DFS/BFS applications), sorting/searching, greedy algorithms. 3. **Math:** Number theory (divisibility, modular arithmetic), combinatorics, probability. 4. **Problem Solving Techniques:** Divide and conquer, binary search, two pointers. 5. **Implementation:** Bit manipulation, recursion, STL.

Type your message here...

Back to Dashboard

10) All users (Leaderboard)

<CodeCase>

Dashboard

All Users

Contests

Problems

Notes

CP Helper

Configure Profiles

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Leaderboard

Users ranked by total problems solved across platforms

Rank

Name

Email

LeetCode

CodeChef

Codeforces

Total

1

omik

omik@omik.com

106

449

283

838

2

Om Thanage

om.thanage@somaiya.edu

34

224

66

324

3

Omkar Dinde

omkar.dinde@somaiya.edu

70

25

47

142

Back to Dashboard

6. Database Schema

Overview

Server 127.0.0.1 > Database: codekeep

Structure

SQL

Search

Query

Export

Import

Operations

Privileges

Routines

Events

Triggers

Designer

Filters

Containing the word:

Table	Action	Rows	Type	Collation	Size	Overhead
☐ bookmarks	★ Browse Structure Search Insert Empty Drop					

 0 | InnoDB | utf8mb4_general_ci | 48.0 KiB | - || ☐ notes | ★ Browse Structure Search Insert Empty Drop 3 | InnoDB | utf8mb4_general_ci | 16.0 KiB | - | |
☐ platform	★ Browse Structure Search insert Empty Drop 3	InnoDB	utf8mb4_general_ci	48.0 KiB	-	
☐ problems	★ Browse Structure Search Insert Empty Drop 5	InnoDB	utf8mb4_general_ci	16.0 KiB	-	
☐ profile	★ Browse Structure Search Insert Empty Drop 1	InnoDB	utf8mb4_general_ci	16.0 KiB	-	
☐ saved_contests	★ Browse Structure Search Insert Empty Drop 1	InnoDB	utf8mb4_general_ci	32.0 KiB	-	
☐ solved_problems	★ Browse Structure Search Insert Empty Drop 0	InnoDB	utf8mb4_general_ci	48.0 KiB	-	
☐ users	★ Browse Structure Search Insert Empty Drop 2	InnoDB	utf8mb4_general_ci	16.0 KiB	-	
☐ user_statistics	★ Browse Structure Search insert Empty Drop 1	InnoDB	utf8mb4_general_ci	32.0 KiB	-	
9 tables	Sum	16	InnoDB	utf8mb4_general_ci	272.0 KiB	0 B

⏮

☐ Check all

With selected:

Users

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	name	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐ 2	email	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐ 3	password	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More

Profile

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	Uemail	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
☐ 2	codechef_id	varchar(255)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 3	codeforces_id	varchar(255)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 4	leetcode_id	varchar(255)	utf8mb4_general_ci		Yes	NULL			Change Drop More

Platform

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	platform_name	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 2	platformId	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐ 3	platformUrl	text	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 4	email	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More

Problems

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐ 2	title	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
☐ 3	difficulty	enum('Easy', 'Medium', 'Hard')	utf8mb4_general_ci		No	None			Change Drop More
☐ 4	platform	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
☐ 5	problem_url	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
☐ 6	description	text	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 7	tags	text	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 8	created_at	timestamp			No	current_timestamp()			Change Drop More

solved_problems

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id 🔑	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2 user_email 🔑	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
☐	3 problem_id 🔑	int(11)			No	None			Change Drop More
☐	4 solution_code	text	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	5 language	varchar(50)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	6 time_taken	int(11)			Yes	NULL			Change Drop More
☐	7 solved_at	timestamp			No	current_timestamp()			Change Drop More

saved_contests

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id 🔑	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2 user_email 🔑	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
☐	3 platform 🔑	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
☐	4 contest_name	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
☐	5 contest_code 🔑	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐	6 contest_start_date	datetime			No	None			Change Drop More
☐	7 contest_end_date	datetime			No	None			Change Drop More
☐	8 created_at	timestamp			No	current_timestamp()			Change Drop More

notes

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id 🔑	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2 user_email	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
☐	3 title	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
☐	4 content	text	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	5 category	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	6 created_at	timestamp			No	current_timestamp()			Change Drop More
☐	7 updated_at	timestamp			No	current_timestamp()		ON UPDATE CURRENT_TIMESTAMP()	Change Drop More

bookmarks

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id 🔑	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2 user_email 🔑	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
☐	3 problem_id 🔑	int(11)			No	None			Change Drop More
☐	4 created_at	timestamp			No	current_timestamp()			Change Drop More

user_statistics

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id 🔑	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2 user_email 🔑	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
☐	3 total_solved	int(11)			Yes	0			Change Drop More
☐	4 easy_solved	int(11)			Yes	0			Change Drop More
☐	5 medium_solved	int(11)			Yes	0			Change Drop More
☐	6 hard_solved	int(11)			Yes	0			Change Drop More
☐	7 leetcode_solved	int(11)			Yes	0			Change Drop More
☐	8 codechef_solved	int(11)			Yes	0			Change Drop More
☐	9 codeforces_solved	int(11)			Yes	0			Change Drop More
☐	10 streak_days	int(11)			Yes	0			Change Drop More
☐	11 last_solved_date	date			Yes	NULL			Change Drop More
☐	12 updated_at	timestamp			No	current_timestamp()		ON UPDATE CURRENT_TIMESTAMP()	Change Drop More

7. Conclusion

CodeCase successfully achieves its goal of creating a unified platform for competitive programmers to manage their coding journey. By integrating with popular platforms like LeetCode, CodeChef, and Codeforces, it provides a centralized hub for:

- Tracking upcoming contests across multiple platforms with IST timezone support
- Managing and categorizing coding problems by difficulty and platform
- Maintaining a personal notes system for algorithms and code snippets
- Receiving AI-assisted guidance through the CP Helper feature
- Monitoring progress through comprehensive statistics and visualization
- The project demonstrates effective use of PHP, MySQL, and modern web technologies to create a responsive and functional tool that addresses the real needs of competitive programmers. With its dark-themed UI optimized for coding and seamless platform integrations, CodeCase streamlines the competitive programming workflow and helps users stay organized in their learning journey.

8. Future scope

1. Extended Platform Integrations: Add support for additional platforms like HackerRank, AtCoder, and SPOJ
2. CP Duels: Implement a feature allowing users to challenge friends to head-to-head competitive programming duels with real-time tracking, custom problem sets, and leaderboards
3. Advanced Analytics: Enhance the statistics module with deeper insights, performance predictions, and personalized improvement recommendations
4. Collaborative Features: Add team formation tools, group study rooms, and collaborative problem-solving capabilities
5. Offline Mode: Enable saving problems and notes for offline access during travel or connectivity issues
6. Custom Contest Creation: Allow users to create private contests with custom problem sets for practice or small group competitions
7. Enhanced AI Assistant: Expand the CP Helper with more advanced algorithm suggestions, code optimization, and personalized learning paths
8. Internationalization: Add support for multiple languages to make the platform accessible to a global audience